Koa Shen

Mechanical Engineer

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Commonwealth Fusion Systems Hiring Team

Re: Summer 2027 Mechanical Engineering and Controls Engineering Internships

Dear Commonwealth Fusion Systems Hiring Team,

I am a mechanical engineering student at UC Santa Barbara, graduating in June 2028, and I am applying for CFS's Summer 2027 mechanical and controls engineering internships. I was referred by Derek Chow, who previously worked at SLAC National Accelerator Laboratory, where I am currently an intern. I am drawn to CFS because building SPARC and its high-temperature superconducting magnets depends on the kind of work I enjoy most: precise hardware, thoughtful tooling and assembly, and motion and control systems that have to work reliably.

At SLAC, I build design and simulation tools for tightly packaged X-ray beamline assemblies. I developed an open-source Python/Drake kinematic model of a 30-DOF, 305-body assembly of motion stages; its millisecond collision checks exposed at least five physical pinch points and more than 10 actionable design changes, including encoder retrofits. I am now working toward EPICS controls compatibility and path planning. I also design mounts, alignment features, and detector strain relief in Solid Edge with GD&T, checking load cases for detectors worth more than $100,000.

As fabrication team lead for UCSB Gaucho Racing, I CNC-machined 33 suspension components from 4130 steel and 7075-T6 aluminum, holding bearing fits within 0.0005 inch, and designed chassis welding fixtures that located 81 tubes within 0.050 inch of CAD while cutting assembly time 50%, then a further 30%. I also led 15 teammates through the design, fabrication, and installation of our driver-safety systems, which passed FSAE technical inspection.

On the controls side, I built a closed-loop actuator: a compact 15:1 cycloidal reducer driven by a stepper motor, with C++ firmware that reads a 12-bit magnetic encoder, logs real-time telemetry, corrects position error, and detects missed steps. I also designed a backlash test that pairs encoder feedback with a limit switch.

I would bring CFS hands-on experience across mechanical design, precision fabrication, fixturing, and closed-loop control, along with a habit of checking models against real hardware. I would be excited to contribute to the tooling, hardware, and control systems that help bring SPARC and commercial fusion power online. Thank you for your consideration.

Sincerely,
Koa Shen